

Affiliate & Referral System — Full Development Assignment

1. Assignment Overview

You are being assigned the development of a complete **Affiliate / Referral Management System** for an e-commerce platform.

The objective is to build a production-ready affiliate system where approved partners can refer customers through unique coupon codes.

The fundamental business model is:

Affiliate/Coupon Holder → Shares Unique Coupon → Customer Uses Coupon → Customer Receives X% Discount → Affiliate Earns Y% Commission

Example

Assume:

- Product/order value: **€200**
- Customer discount: **10%**
- Affiliate commission: **5%**

Calculation:

- Customer discount = €20
- Customer pays = €180
- Affiliate commission = €9, assuming commission is calculated on the discounted eligible value.

The system must keep the **customer discount and affiliate commission completely separate**.

2. Assignment Objective

You are expected to design and implement the complete system, including:

1. Affiliate registration
2. Affiliate approval
3. Affiliate profiles
4. Affiliate dashboard
5. Coupon creation and management
6. Coupon validation

7. Customer discount calculation
8. Referral attribution
9. Order attribution
10. Commission calculation
11. Commission ledger
12. Commission status management
13. Refund/reversal handling
14. Affiliate payout management
15. Admin affiliate management
16. Admin reporting
17. Fraud prevention
18. Notifications
19. Audit logs
20. Security and access control
21. Database architecture
22. API architecture
23. Edge-case handling
24. Testing

The implementation should be structured so that the system can later support additional affiliate functionality without requiring a complete rewrite.

3. User Roles

The system must support three primary roles.

3.1 Customer

The customer is the person purchasing products.

The customer can:

- Enter an affiliate coupon
- Apply a coupon
- Receive the configured discount
- See the discount during checkout
- Complete payment
- Receive order confirmation

The customer **must not see the affiliate's commission.**

3.2 Affiliate / Coupon Holder

The affiliate is the person or business referring customers.

The affiliate can:

- Register
 - Submit an application
 - Receive approval
 - Access an affiliate dashboard
 - Receive a unique coupon
 - Share the coupon
 - View referred orders
 - View sales generated
 - View commissions
 - View commission status
 - View payout history
-

3.3 Administrator

The administrator controls the complete system.

Admin can:

- Create affiliates
 - Approve affiliates
 - Reject affiliates
 - Suspend affiliates
 - Create coupons
 - Assign coupons
 - Set customer discount
 - Set affiliate commission
 - Configure commission calculation
 - View referrals
 - View orders
 - View commissions
 - Approve commissions
 - Reject commissions
 - Reverse commissions
 - Manage payouts
 - Generate reports
 - Export data
 - Investigate suspicious activity
-

4. Affiliate Registration

Create an affiliate registration/application system.

Required fields

Personal Information

- Full Name
- Email
- Phone
- Country
- Address

Professional Information

- Business Name
- Business Type
- Website
- Social Media Profiles
- Audience Type
- Description
- Expected Referral Volume

Payment Information

Support the required payout information, including:

- Account Holder Name
- Bank Name
- Account Number
- IBAN
- BIC/SWIFT
- Payment Method
- Billing Information

Financial information must be securely stored and must not be exposed to unauthorized users.

5. Affiliate Approval Workflow

Affiliate registration must not automatically activate the affiliate.

Workflow

Application Submitted
↓
Pending
↓
Admin Review
↓
Approved / Rejected

↓
Approved
↓
Active

Possible statuses:

- Pending
- Approved
- Active
- Rejected
- Suspended
- Deactivated

Only **Active** affiliates can generate valid referrals and commissions.

6. Affiliate ID

Every affiliate must receive a unique internal identifier.

Example:

AFF-000001
AFF-000002
AFF-000003

The Affiliate ID must remain permanent even if the affiliate changes their name, coupon or other profile information.

7. Affiliate Profile

Each affiliate profile must contain:

- Affiliate ID
- Name
- Email
- Phone
- Business Name
- Country
- Status
- Registration Date
- Approval Date
- Coupon Code
- Customer Discount %
- Commission %

- Commission Calculation Base
 - Total Referrals
 - Total Orders
 - Total Sales
 - Total Discounts
 - Total Commission
 - Pending Commission
 - Approved Commission
 - Paid Commission
 - Cancelled Commission
 - Reversed Commission
-

8. Coupon System

Each affiliate should be able to have a unique coupon code.

Example:

Affiliate:
AFF-000123

Coupon:
ALEX10

The coupon must be linked to the affiliate.

Affiliate
↓
Affiliate Coupon
↓
Customer
↓
Order
↓
Commission

9. Coupon Configuration

Each coupon must support:

- Coupon ID
- Coupon Code
- Affiliate ID
- Discount Type
- Discount Value
- Commission Type
- Commission Value

- Start Date
 - Expiry Date
 - Minimum Order Value
 - Maximum Discount
 - Global Usage Limit
 - Per Customer Usage Limit
 - Eligible Products
 - Excluded Products
 - Eligible Categories
 - Excluded Categories
 - Active/Inactive Status
-

10. Coupon Uniqueness

Coupon codes must be globally unique.

For example:

```
ALEX10  
alex10  
Alex10
```

must not be treated as three separate active coupon codes.

The backend must normalise coupon codes before validation.

The database must enforce uniqueness.

11. Customer Discount

The affiliate coupon determines the customer's discount.

Example:

```
Coupon: ALEX10  
Customer Discount: 10%  
Commission: 5%
```

Customer enters:

```
ALEX10
```

The system calculates:

```
Eligible Value = €200
```

10% Discount = €20

Customer Payable = €180

Checkout should display:

Subtotal	€200.00
Affiliate Discount 10%	-€20.00

Total	€180.00

12. Affiliate Commission

Affiliate commission is a separate calculation.

Using the same example:

Eligible Value = €200
Customer Discount = €20

Commission Base = €180

Commission Rate = 5%

Affiliate Commission = €9

The customer receives:

€20 discount

The affiliate receives:

€9 commission

The customer must never see the €9 affiliate commission.

13. X% and Y% Must Be Independent

This is a critical requirement.

The system must support:

Customer Discount = X%
Affiliate Commission = Y%

For example:

Discount = 15%
Commission = 7%

or:

Discount = 10%
Commission = 5%

or:

Discount = 20%
Commission = 10%

Changing the customer discount must not automatically change the commission.

Changing the commission must not automatically change the customer discount.

14. Commission Calculation Base

The developer must implement a configurable commission calculation base.

Admin should be able to choose:

Option 1

Commission based on original eligible product value.

Option 2

Commission based on discounted product value.

Option 3

Commission based on product value excluding tax.

Option 4

Commission based on product value excluding tax and shipping.

The selected rule must be stored with the commission record.

Recommended default

Discounted eligible product value excluding shipping.

15. Referral Attribution

The primary referral attribution mechanism is the affiliate coupon.

Example:

```
Customer
↓
Enters ALEX10
↓
Backend identifies:
AFF-000123
↓
Customer completes order
↓
Order linked to AFF-000123
```

The affiliate relationship must be permanently stored with the order.

16. Order Attribution Snapshot

When an order is created, store:

```
affiliate_id
affiliate_coupon_id
affiliate_coupon_code
customer_discount_rate
customer_discount_amount
commission_rate
commission_base_type
commission_base_amount
commission_amount
```

Do not dynamically calculate historical commission using the affiliate's current configuration.

17. Historical Commission Protection

Example:

Today:

```
ALEX10
Discount = 10%
Commission = 5%
```

Order #1001 uses ALEX10.

Tomorrow admin changes:

Discount = 15%
Commission = 8%

Order #1001 must continue to show:

Discount = 10%
Commission = 5%

New orders use:

Discount = 15%
Commission = 8%

This is mandatory.

18. Coupon Validation

When a customer enters a coupon, backend must verify:

1. Coupon exists
2. Coupon is active
3. Coupon has started
4. Coupon has not expired
5. Affiliate exists
6. Affiliate is active
7. Global usage limit
8. Customer usage limit
9. Minimum order value
10. Product eligibility
11. Category eligibility
12. Excluded products
13. Coupon stacking rules
14. Customer eligibility
15. Order eligibility

Only after successful validation should the discount be applied.

19. Server-Side Financial Calculations

All financial calculations must happen on the backend.

Never trust:

- Discount amount sent from frontend
- Commission amount sent from frontend
- Product price sent from frontend
- Final total sent from frontend

The backend must independently calculate:

```
Product Price
↓
Eligible Value
↓
Discount
↓
Net Value
↓
Tax/Shipping according to configuration
↓
Final Order
↓
Commission
```

20. Coupon Application Flow

```
Customer enters coupon
↓
Frontend sends coupon to API
↓
Backend validates coupon
↓
Identify affiliate
↓
Validate affiliate status
↓
Calculate eligible items
↓
Calculate customer discount
↓
Calculate order total
↓
Return validated result
↓
Customer completes checkout
↓
Order created
↓
Affiliate attribution stored
↓
Payment processed
↓
Commission created
```

21. Commission Creation

Recommended workflow:

```
Order Created
  ↓
Payment Successful
  ↓
Commission Created
  ↓
PENDING
  ↓
Return/Cancellation Period
  ↓
APPROVED
  ↓
Payout
  ↓
PAID
```

Commission should not become payable immediately unless the business explicitly chooses that behaviour.

22. Commission Statuses

Implement:

PENDING

Commission has been generated but is not yet payable.

APPROVED

Commission has passed the required validation period.

PAID

Commission has been included in a completed payout.

REJECTED

Commission rejected by admin.

CANCELLED

Associated order was cancelled.

REVERSED

Commission was previously generated/approved but must be reversed due to refund, chargeback or another event.

23. Order Cancellation

Example:

Order = €200
Commission = €9

If the order is cancelled:

Order → Cancelled
Commission → Cancelled

No payout should be made.

24. Full Refund

If an order is fully refunded:

Order → Refunded
Commission → Reversed

If the commission was already paid, create a negative adjustment/reversal entry rather than deleting the original transaction.

25. Partial Refund

Example:

Original eligible value = €200
Commission = 5%
Commission = €10

Refunded eligible value = €80

Commission reversal:

$€80 \times 5\% = €4$

Remaining commission:

$€10 - €4 = €6$

The system must automatically support proportional commission reversal.

26. Multiple Affiliate Coupons

Default rule:

Only one affiliate coupon may be applied to a single order.

The system must prevent:

ALEX10
+
JOHN10

from generating commissions for two affiliates.

Admin may configure replacement behaviour if required.

27. Affiliate Coupon + Other Discounts

Admin must be able to configure whether affiliate coupons can be combined with:

- Product discounts
- Sale prices
- Promotional coupons
- Other coupons
- B2B discounts

Setting:

Allow Coupon Stacking: YES / NO

Recommended default:

NO

28. Self-Referral Prevention

Affiliates should not earn commission from their own purchases.

The system should flag possible self-referrals using relevant account information.

Potential matching data:

- Affiliate email
- Customer email
- Phone
- Account identity
- Billing details
- Shipping details

The system should allow admin review/override.

29. Affiliate Dashboard

Create a dedicated affiliate dashboard.

Dashboard Overview

Display:

- Total referrals
 - Total referred customers
 - Total orders
 - Total sales
 - Total customer discounts
 - Total commission
 - Pending commission
 - Approved commission
 - Paid commission
-

30. Affiliate Coupon Section

Example:

MY AFFILIATE COUPON

Coupon Code:
ALEX10

Customer Discount:
10%

My Commission:
5%

Status:
ACTIVE

Affiliate should have a convenient way to copy the coupon code.

31. Affiliate Orders

Display:

Order ID	Date	Order Value	Discount	Commission	Status
#1001	31/08/26	€200	€20	€9	Approved
#1002	02/09/26	€150	€15	€6.75	Pending

Do not expose unnecessary customer personal information.

32. Affiliate Commission Page

Display:

- Commission ID
 - Order ID
 - Order date
 - Commission base
 - Commission rate
 - Commission amount
 - Status
 - Approval date
 - Payment date
-

33. Affiliate Payout Page

Display:

- Total earned
- Pending
- Approved
- Paid
- Available for payout
- Minimum payout threshold
- Payout history

Example:

Total Commission	€720
Pending	€120
Approved	€200

Paid	€400
Available	€200

34. Admin Affiliate Dashboard

Admin dashboard should provide:

KPIs

- Total Affiliates
 - Active Affiliates
 - Pending Affiliates
 - Suspended Affiliates
 - Total Referral Orders
 - Total Referred Revenue
 - Total Customer Discounts
 - Total Commission
 - Pending Commission
 - Approved Commission
 - Paid Commission
-

35. Affiliate Management

Admin should be able to:

- Search affiliates
- Filter affiliates
- View affiliate
- Edit affiliate
- Approve
- Reject
- Suspend
- Reactivate
- Deactivate

Filters:

- Status
 - Date
 - Sales
 - Commission
 - Coupon
 - Country
-

36. Create Affiliate

Admin form:

Full Name
Email
Phone
Business Name
Country
Status

Coupon Code

Customer Discount %
Commission %

Commission Base

Start Date
Expiry Date

Minimum Order Value
Maximum Discount
Usage Limit
Per Customer Limit

Eligible Products
Excluded Products
Eligible Categories
Excluded Categories

Coupon Stacking

37. Commission Ledger

Do not calculate historical commission totals only from current order data.

Create a dedicated commission ledger.

Each commission record must contain:

Commission ID
Affiliate ID
Coupon ID
Coupon Code
Order ID
Customer ID/reference
Order Value
Eligible Value
Discount Amount
Commission Base
Commission Rate
Commission Amount
Currency

Status
Created At
Approved At
Paid At
Reversed At
Reversal Amount
Notes

38. Payout Architecture

Admin should be able to create a payout.

Example:

Affiliate:
AFF-000123

Approved Commission:
€720

Already Paid:
€500

Available:
€220

Admin can:

- Select commissions
 - Create payout
 - Enter payout amount
 - Enter payment method
 - Enter payment reference
 - Enter payment date
 - Upload payment proof if required
 - Add notes
 - Mark as paid
-

39. Payout Status

PENDING
↓
PROCESSING
↓
PAID

or:

PENDING

↓
REJECTED

40. Payout Threshold

Admin should be able to configure a minimum payout threshold.

Example:

Minimum Payout Threshold = €50

Affiliate has:

€37

Dashboard should show:

€13 remaining to reach the payout threshold.

The threshold must be configurable.

41. Payout Frequency

System architecture should support:

- Manual
- Weekly
- Bi-weekly
- Monthly

Initial implementation may use:

Manual admin-controlled payouts

The architecture should allow automated payouts to be added later.

42. Payout Ledger

Every payout must have a permanent record.

Payout

Payout ID

Affiliate ID
Gross Amount
Adjustments
Net Amount
Currency
Payment Method
Payment Date
Payment Reference
Status
Notes
Created At

Payout Items

Payout ID
Commission ID
Amount

This creates a clear relationship:

Affiliate
↓
Commission Records
↓
Payout
↓
Payout Items

43. Fraud Detection

The system should flag:

- Self-referrals
- Suspicious repeated orders
- Excessive coupon usage
- Multiple accounts
- Unusual commission spikes
- High cancellation rate
- High refund rate
- Abnormal order behaviour
- Coupon abuse

Admin should be able to:

- Flag affiliate
 - Hold commissions
 - Suspend affiliate
 - Reject commission
 - Investigate order
 - Add internal notes
-

44. Commission Holding

Admin should be able to place commissions on hold.

Example:

Commission:
€250

Status:
ON HOLD

Reason:
Fraud Review

The hold must prevent payout without deleting the commission record.

45. Reporting

Affiliate Sales Report

Include:

- Affiliate
- Coupon
- Orders
- Gross sales
- Discounts
- Net sales
- Commission

Commission Report

Include:

- Affiliate
- Order
- Commission
- Status
- Date

Coupon Performance

Include:

- Coupon usage

- Orders
 - Revenue
 - Customer discounts
 - Commission generated
-

46. Profitability Reporting

The admin system should be capable of calculating:

```
Gross Sales
-
Customer Discounts
-
Affiliate Commissions
-
Refunds
=
Affiliate-attributed Net Revenue
```

This is important for understanding the actual cost of the affiliate programme.

47. Date Filtering

Reports must support:

- Today
 - Yesterday
 - Last 7 Days
 - Last 30 Days
 - This Month
 - Previous Month
 - Custom Date Range
-

48. Export

Admin should be able to export:

- Affiliate list
- Coupon list
- Referral orders
- Commission ledger
- Payout ledger

- Affiliate performance
- Sales reports

Preferred formats:

- CSV
 - Excel
 - PDF
-

49. Notification System

Affiliate should receive notifications for:

Registration

Affiliate application received.

Approval

Your affiliate account has been approved.

Coupon Activation

Your affiliate coupon is now active.

Referral Order

A customer has placed an order using your coupon.

Commission

Commission has been generated.

Approval

Your commission has been approved.

Payout

Your affiliate commission has been paid.

Suspension

Your affiliate account has been suspended.

50. Customer Checkout Experience

Checkout should contain:

Have a referral coupon?

[ALEX10] [APPLY]

After applying:

✓ Coupon ALEX10 applied

Affiliate Discount: 10%

You saved: €20.00

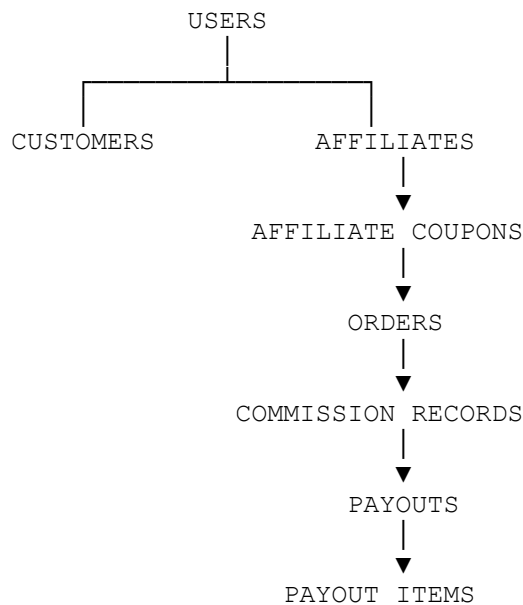
The customer must **not** see:

Affiliate Commission: €9

Commission information is restricted to the affiliate and authorised administrators.

51. Database Architecture

Recommended architecture:



52. affiliates Table

Suggested fields:

```
id
affiliate_code
user_id
name
email
phone
business_name
country
address
status
commission_rate
commission_base_type
created_at
approved_at
updated_at
```

53. **affiliate_coupons** Table

```
id
affiliate_id
coupon_code
discount_type
discount_value
commission_rate
commission_base_type
minimum_order_value
maximum_discount
usage_limit
per_customer_limit
start_at
expires_at
status
created_at
updated_at
```

54. **affiliate_referrals** Table

```
id
affiliate_id
customer_id
coupon_id
first_attributed_at
last_attributed_at
status
```

55. **orders** Table

Add affiliate-related fields:

```
affiliate_id
affiliate_coupon_id
```

affiliate_coupon_code
affiliate_discount_amount
affiliate_commission_base
affiliate_commission_rate
affiliate_commission_amount

56. `commission_records` Table

id
affiliate_id
order_id
coupon_id
commission_base
commission_rate
commission_amount
currency
status
created_at
approved_at
paid_at
reversed_at
reversal_amount
notes

57. `payouts` Table

id
affiliate_id
payout_reference
gross_amount
adjustment_amount
net_amount
currency
status
payment_method
payment_date
payment_reference
notes
created_at

58. `payout_items` Table

id
payout_id
commission_id
amount

59. API Architecture

The system should have separate APIs for:

Affiliate

```
POST    /affiliate/register
GET     /affiliate/profile
GET     /affiliate/dashboard
GET     /affiliate/coupon
GET     /affiliate/orders
GET     /affiliate/commissions
GET     /affiliate/payouts
```

Coupon

```
POST    /coupon/validate
POST    /coupon/apply
DELETE  /coupon/remove
```

Admin

```
GET     /admin/affiliates
POST    /admin/affiliates
GET     /admin/affiliates/:id
PUT     /admin/affiliates/:id
POST    /admin/affiliates/:id/approve
POST    /admin/affiliates/:id/suspend

GET     /admin/coupons
POST    /admin/coupons
PUT     /admin/coupons/:id

GET     /admin/commissions
PUT     /admin/commissions/:id
POST    /admin/commissions/:id/approve
POST    /admin/commissions/:id/reverse

GET     /admin/payouts
POST    /admin/payouts
POST    /admin/payouts/:id/mark-paid
```

These are architectural examples; adapt naming to the project's actual API conventions.

60. API Security

Every API must enforce:

- Authentication
- Authorisation
- Role validation
- Input validation
- Rate limiting where appropriate
- Server-side calculations

- Ownership checks

An affiliate must never be able to request another affiliate's:

- Orders
- Customers
- Commissions
- Payouts
- Financial information

by changing an ID in an API request.

61. Duplicate Commission Protection

This is a mandatory requirement.

Payment providers may send the same webhook multiple times.

Example:

```
Payment Webhook
  ↓
Webhook #1
  ↓
Commission Created

Webhook #2
  ↓
Existing Commission Found
  ↓
DO NOT CREATE AGAIN
```

The commission creation process must be **idempotent**.

Use appropriate unique constraints/idempotency keys.

62. Financial Precision

All money calculations must use decimal-safe monetary handling.

Do not use unsafe floating-point calculations for financial values.

Example:

```
€200.00 × 10% = €20.00
€180.00 × 5% = €9.00
```

Currency must be stored for every financial transaction.

63. Tax Architecture

The commission calculation must distinguish between:

Product Value
Discount
Tax
Shipping
Commission Base
Commission

Do not simply calculate commission from the final order total.

The system should allow configuration of whether commission is calculated:

- Before tax
 - After tax
 - Before shipping
 - After shipping
 - Before discount
 - After discount
-

64. Audit Log

The following actions must be logged:

- Affiliate created
- Affiliate updated
- Affiliate approved
- Affiliate rejected
- Affiliate suspended
- Affiliate reactivated
- Coupon created
- Coupon modified
- Coupon disabled
- Commission created
- Commission approved
- Commission rejected
- Commission cancelled
- Commission reversed
- Commission placed on hold
- Payout created
- Payout modified

- Payout marked paid

Audit log should contain:

User
Action
Entity
Entity ID
Old Value
New Value
Timestamp
IP / Session information where appropriate

65. Security Requirements

The implementation must include:

- Role-based access control
 - Secure authentication
 - Server-side validation
 - Secure coupon validation
 - Server-side financial calculations
 - Database constraints
 - Secure payment handling
 - Protection against duplicate commissions
 - Protection against coupon manipulation
 - Protection against price manipulation
 - Protection against unauthorised data access
 - Audit logging
 - Secure storage of payout information
-

66. Edge Cases to Handle

The implementation must explicitly handle the following.

Case 1 — Valid Coupon

Customer uses valid coupon.

Expected:

Discount applied + affiliate attribution + commission generated after successful payment.

Case 2 — Expired Coupon

Expected:

Coupon rejected.

No commission.

Case 3 — Suspended Affiliate

Expected:

Coupon becomes invalid for new transactions.

Case 4 — Usage Limit Reached

Expected:

Coupon rejected.

Case 5 — Customer Limit Reached

Expected:

Coupon rejected for that customer.

Case 6 — Order Cancelled

Expected:

Commission cancelled.

Case 7 — Full Refund

Expected:

Commission reversed.

Case 8 — Partial Refund

Expected:

Corresponding portion of commission reversed.

Case 9 — Commission Rate Changed

Old orders retain the old rate.

New orders use the new rate.

Case 10 — Coupon Disabled

Historical orders remain associated with the coupon.

Case 11 — Two Affiliate Coupons

Only one affiliate can receive commission for the order.

Case 12 — Failed Payment

No payable commission should be created.

Case 13 — Duplicate Payment Webhook

Only one commission should exist.

Case 14 — Self Referral

Commission must be blocked or flagged according to configured rules.

Case 15 — Affiliate Suspended After Existing Order

Existing historical commissions remain traceable.

New referrals must be blocked.

67. Testing Requirements

You must test:

Functional Testing

- Registration
- Login
- Approval
- Coupon creation
- Coupon validation
- Discount calculation
- Commission calculation
- Referral attribution
- Order attribution
- Refund handling
- Payout handling

Security Testing

- Role access
- API authorisation
- ID manipulation
- Coupon manipulation
- Price manipulation
- Commission manipulation

Financial Testing

Test:

- 1%
- 5%
- 10%
- 15%
- 25%
- Decimal amounts
- Large orders
- Small orders
- Zero-value edge cases
- Refunds
- Partial refunds

Concurrency Testing

Test:

- Two customers using same coupon simultaneously
 - Coupon usage limits
 - Duplicate payment callbacks
 - Duplicate order requests
-

68. Acceptance Criteria

The assignment will be considered complete only when:

Affiliate

- Affiliate can register.
- Admin can approve affiliate.
- Affiliate receives unique coupon.
- Affiliate can view dashboard.

Customer

- Customer can enter coupon.
- Coupon validates correctly.
- Customer receives X% discount.
- Customer does not see affiliate commission.

Commission

- Affiliate receives Y% commission.
- X and Y are independently configurable.
- Commission base is configurable.
- Commission is stored as a snapshot.
- Commission cannot be duplicated.
- Commission reacts correctly to cancellation/refund.

Admin

- Admin can manage affiliates.
- Admin can manage coupons.
- Admin can manage commission.
- Admin can manage payouts.
- Admin can view reports.
- Admin can export records.

Security

- Affiliate cannot access another affiliate's information.

- Customer cannot access affiliate information.
 - Financial calculations cannot be manipulated from frontend.
 - Duplicate commission generation is prevented.
-

69. Required Deliverables From Applicant

The applicant must submit:

1. System Architecture

Provide:

- Architecture diagram
- Component structure
- Data flow
- Authentication flow
- Affiliate flow
- Coupon flow
- Commission flow
- Payout flow

2. Database Design

Provide:

- ER diagram
- Tables
- Relationships
- Primary keys
- Foreign keys
- Indexes
- Unique constraints

3. API Documentation

Provide:

- Endpoint
- Method
- Request
- Response
- Authentication
- Validation
- Error handling

4. UI/UX

Provide designs/screens for:

- Affiliate registration
- Affiliate dashboard
- Coupon section
- Orders
- Commission
- Payouts
- Admin affiliate management
- Admin coupon management
- Admin commission management
- Admin payout management

5. Working Prototype

The applicant must demonstrate:

```
Affiliate Registration
↓
Admin Approval
↓
Coupon Creation
↓
Customer Uses Coupon
↓
Customer Gets X% Discount
↓
Order Completed
↓
Affiliate Gets Y% Commission
↓
Commission Pending
↓
Commission Approved
↓
Payout
↓
Commission Paid
```

70. Mandatory Demonstration Scenario

Every applicant must demonstrate the following exact scenario.

Configuration

Affiliate:
Alex

Coupon:
ALEX10

Customer Discount:
10%

Affiliate Commission:
5%

Commission Base:
Discounted Product Value

Minimum Payout:
€50

Customer Order

Product Value = €200

Customer applies:

ALEX10

System must calculate:

Original Value:	€200.00
Customer Discount:	-€20.00
Customer Pays:	€180.00

Affiliate commission:

Commission Base:	€180.00
Commission Rate:	5%
Commission:	€9.00

Then demonstrate:

Payment Successful
↓
Commission = Pending
↓
Order Completed / Return Period Passed
↓
Commission = Approved
↓
Payout Created
↓
Commission = Paid

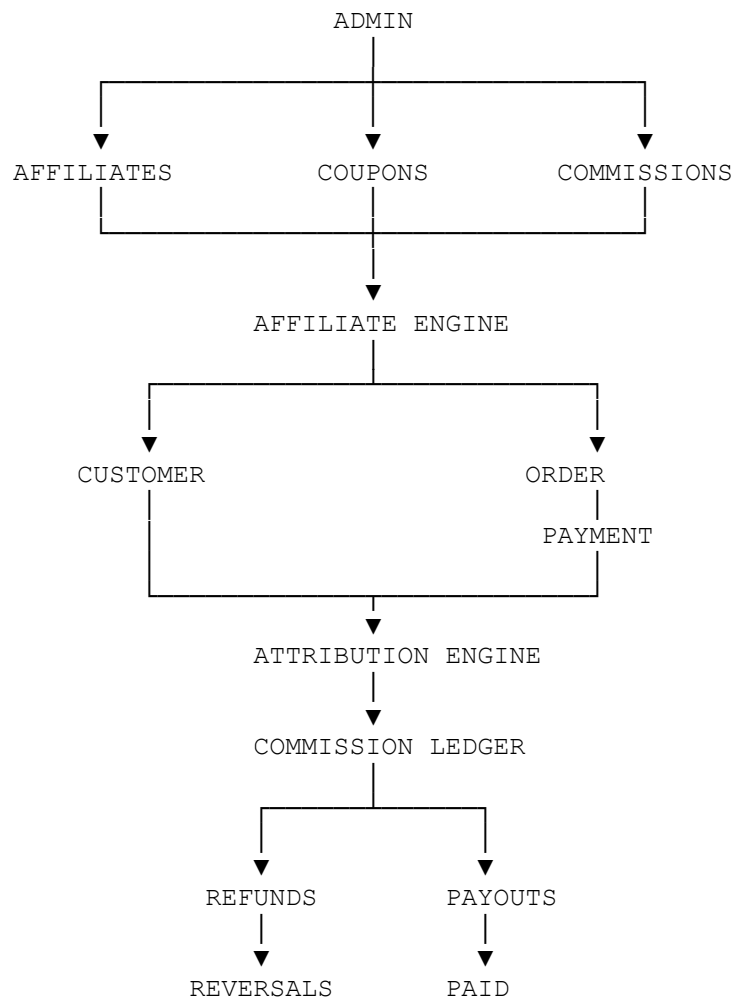
Then demonstrate:

Customer Refunds €80
↓
Commission Reversal = €4
↓
Remaining Commission = €5

The applicant must show that the complete transaction remains auditable.

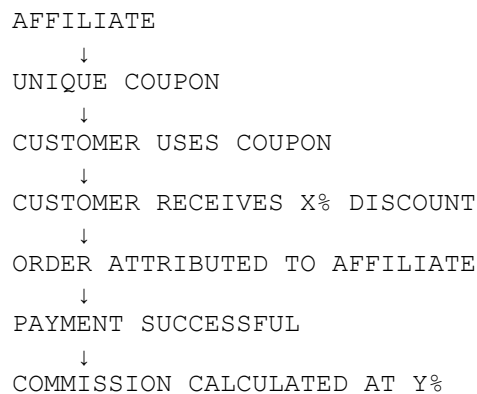
71. Expected Final Architecture

The final system should conceptually operate as:



72. Core Business Rule

The entire implementation must ultimately follow this rule:



↓
COMMISSION PENDING
↓
REFUND / CANCELLATION PERIOD
↓
COMMISSION APPROVED
↓
PAYOUT THRESHOLD REACHED
↓
PAYOUT PROCESSED
↓
COMMISSION PAID

Non-Negotiable Principle

The customer discount and affiliate commission are two independent financial rules.

CUSTOMER BENEFIT
X% DISCOUNT

≠

AFFILIATE BENEFIT
Y% COMMISSION

Both values must be independently configurable, independently calculated, independently stored, independently reported, and independently audited.